Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 5 - 09/22 Content

MATH 1271

09/24/2026

This quiz is worth 5 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	5
Total:	5

1. (5 points) If $f(x) = \dfrac{2}{x^2}$, find $f'(3)$ using the definition of the derivative.

Solution:

$$f'(3) = \lim_{x\to 3} \frac{f(x) - f(3)}{x - 3} \tag{1}$$

$$= \lim_{x\to 3} \frac{\frac{2}{x^2} - \frac{2}{3^2}}{x - 3} \tag{2}$$

$$= \lim_{x\to 3} \frac{x^2}{x^2} \cdot \frac{\frac{2}{x^2} - \frac{2}{9}}{x - 3} \tag{3}$$

$$= \lim_{x\to 3} \frac{2}{x^2} \cdot \frac{1 - \frac{x^2}{9}}{x - 3} \tag{4}$$

$$= \lim_{x\to 3} \frac{2}{3x^2} \cdot \frac{(1 - \frac{x}{3})(1 + \frac{x}{3})}{\frac{x}{3} - 1} \tag{5}$$

$$= \lim_{x\to 3} -\frac{2}{3x^2} \frac{(1 - \frac{x}{3})(1 + \frac{x}{3})}{1 - \frac{x}{3}} \tag{6}$$

$$= \lim_{x\to 3} -\frac{2}{3x^2}(1 + \frac{x}{3}) = -\frac{2}{3 \cdot 3^2}(1 + \frac{3}{3}) = -\frac{4}{27} \tag{7}$$

Or, alternatively:

$$f'(3) = \lim_{h\to 0} \frac{f(3 + h) - f(3)}{h} \tag{8}$$

$$= \lim_{h\to 0} \frac{\frac{2}{(3+h)^2} - \frac{2}{9}}{h} \tag{9}$$

$$= \lim_{h\to 0} \frac{(3 + h)^2}{(3 + h)^2} \cdot \frac{\frac{2}{(3+h)^2} - \frac{2}{9}}{h} \tag{10}$$

$$= \lim_{h\to 0} \frac{2}{(3 + h)^2} \cdot \frac{1 - \frac{(3+h)^2}{9}}{h} \tag{11}$$

$$= \lim_{h\to 0} \frac{2}{(3 + h)^2} \cdot \frac{1 - \frac{9+6h+h^2}{9}}{h} \tag{12}$$

$$= \lim_{h\to 0} \frac{2}{(3 + h)^2} \cdot \frac{1 - \frac{9}{9} - \frac{6h+h^2}{9}}{h} \tag{13}$$

$$= \lim_{h\to 0} -\frac{2}{(3 + h)^2} \cdot \frac{6h + h^2}{9h} \tag{14}$$

$$= \lim_{h\to 0} -\frac{2}{(3 + h)^2} \cdot \frac{6 + h}{9} \tag{15}$$

$$= -\frac{2}{(3 + 0)^2} \cdot \frac{6 + 0}{9} = -\frac{2}{9} \cdot \frac{6}{9} = -\frac{4}{27} \tag{16}$$